

PROJECT EXECUTABLE FILES

EduGenie: Google Gemini Powered Learning Assistant

Field	Details
Team Members	A. Mehathaj Banu S. Haridharshini D. Yogalakshmi D. Hemavathy S V Malini
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Project ID	6ab7f80ad083b0e2ee72b17e
Date	2 October 2026
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Status	Remarks
1	Complete source code (all files and folders)	Yes	FastAPI backend, feature modules, HTML/CSS UI, tests and configuration are included in the supplied project content.
2	README / Setup Guide	Yes	README provides VS Code setup, virtual environment, dependency installation, .env configuration, Uvicorn startup, API endpoints and testing instructions.
3	requirements.txt / package.json / dependency file	Yes	requirements.txt content is provided with FastAPI, Uvicorn, Jinja2, google-genai, python-dotenv and Pydantic; optional torch/transformers dependencies are also specified.
4	Database schema / seed files (if applicable)	NA	The project specifies no database and no user data storage. Local template files are used.
5	Environment configuration file (.env.example or similar)	Yes	.env is expected for GEMINI_API_KEY; README instructs copying .env.example to .env and not sharing the secret.
6	Deployed application URL (if hosted)	NA	No hosted/deployed URL is supplied in the project content. The README specifies local access at http://127.0.0.1:8000 .
7	APK / Executable binary (if applicable)	NA	This is a web application using FastAPI, HTML/CSS and Jinja2; no mobile/desktop binary is specified.
8	Dockerfile / Containerization config (if applicable)	NA	Docker is listed as a recommended future improvement, not as a supplied deployment file.
9	Test files and test results	Yes	Test.quiz validates quiz

			JSON cleanup and response structure without paid API calls. README also provides the unittest command.
10	Demo video or walkthrough (if required)	Not provided	No demo video/link is included in the supplied content.

Step 2: File / Folder Structure

The following structure is derived from the supplied program, README and requirements content.

```
EduGenie/
├── main.py
├── ai_client.py
├── qna.py
├── explanation_module.py
├── quiz_module.py
├── summary_module.py
├── learning_path.py
├── requirements.txt
├── README.md
├── .env.example
├── tests/
│   └── test_quiz.py
├── static/
│   └── style.css
└── templates/
    └── index.html
```

File / Folder	Purpose
main.py	FastAPI application entry point, routes, request validation, static files, Jinja2 templates and health endpoint.
ai_client.py	Shared Google Gemini client; loads GEMINI_API_KEY and generates model responses.
qna.py	Handles learner question answering through Gemini.
explanation_module.py	Concept explanation using optional local LaMini-Flan-T5-783M with Gemini fallback.
quiz_module.py	Generates exactly three MCQ questions and validates/parses JSON output.
summary_module.py	Summarizes learner-provided study text.
learning_path.py	Creates an ordered learning path with topics, time estimates, projects and resource types.
static/style.css	Responsive user interface styling.
templates/index.html	EduGenie web interface and client-side request/result handling.
tests/test_quiz.py	Unit tests for quiz JSON cleanup.
README.md	Setup, use, API, local model, testing and project notes.
requirements.txt	Python dependencies and optional local-model dependencies.
.env.example	Environment-variable template for Gemini configuration; secret values should not be committed.

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not provided. Local URL: http://127.0.0.1:8000

Login Credentials (Demo)	Not applicable / not provided. No authentication flow is specified.
Platform / Hosting Provider	Local development using FastAPI + Uvicorn. Cloud deployment is listed as a recommended future improvement.
Repository Link	Not provided in the supplied content.
Demo Video Link	Not provided in the supplied content.

Application/API access:

- Web application: <http://127.0.0.1:8000>
- FastAPI documentation: <http://127.0.0.1:8000/docs>
- Health check: GET /health
- Feature endpoints: POST /qa, /explain, /quiz, /summarize, /learn/recommendations

Step 4: Run Instructions

1. Extract the EduGenie project and open the project folder in VS Code.
2. Open Terminal → New Terminal.
3. Create a virtual environment: `py -m venv .venv` (Windows) or `python3 -m venv .venv` (macOS/Linux).
4. Activate the environment: `.venv\Scripts\Activate.ps1` on Windows PowerShell, or `source .venv/bin/activate` on macOS/Linux.
5. Install dependencies: `python -m pip install --upgrade pip`
6. Install project dependencies: `pip install -r requirements.txt`
7. Create `.env` from `.env.example` and add the Gemini API key as `GEMINI_API_KEY`. Do not commit or share `.env`.
8. Start the application: `uvicorn main:app --reload`
9. Open <http://127.0.0.1:8000> in a browser.
10. For API testing, open <http://127.0.0.1:8000/docs>.
11. To run the supplied unit tests: `python -m unittest discover -s tests -v`

Optional local explanation model:

Install torch and transformers if `USE_LOCAL_EXPLAINER=true` is required. The README states that the LaMini-Flan-T5-783M model downloads on first use, runs on CPU, and may require substantial disk/RAM. If it fails and `LOCAL_FALLBACK_TO_GEMINI=true`, the application can fall back to Gemini.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	AI output may be inaccurate.	Verify important information; the UI also displays an AI-study-aid disclaimer.
2	Gemini-powered features require internet access and a valid GEMINI_API_KEY.	Configure the <code>.env</code> file correctly. The optional local explanation model can reduce network dependence for

		/explain.
3	The local LaMini-Flan-T5-783M model is optional and can be slow/resource-intensive on CPU.	Keep USE_LOCAL_EXPLAINER=false to use Gemini by default, or provide sufficient CPU/RAM and the required dependencies.
4	No user data/database storage is implemented.	Use the application as a session-based learning assistant; persistent progress tracking would require a future storage layer.
5	No authentication is specified.	Authentication can be added as a future security improvement.
6	No hosted deployment URL, repository link or demo video link is supplied.	Add these links if the submission platform requires them.
7	The supplied source snippets should be syntax/indentation-checked before final execution.	Run the application and unittest suite in VS Code before submission; correct any formatting issues introduced while copying the snippets.
8	Cloud deployment, HTTPS, rate limiting, health checks, retry/fallback, caching, Docker and load balancing are listed as recommended improvements rather than supplied deployment features.	Treat these as future enhancements unless implemented before submission.

Project Content Mapping / Remarks

The broader project material supplied by the team is mapped to the executable-files template as follows:

Area	Content
Project purpose	EduGenie is a Gemini-powered adaptive learning assistant for explanation, Q&A, quiz generation, summarization and learning recommendations.
Functional requirements	FR-1 Diagnostic & Gap Detection; FR-2 Study Plan Generation; FR-3 Multi-Format Learning; FR-4 Progress & Goal Monitoring; FR-5 Context-Aware Support.
Technical architecture	Student → HTML/CSS UI → FastAPI → Feature Module → AI Model → Response → Jinja2 → UI.
AI models	Local: LaMini-Flan-T5-783M for optional /explain. Cloud: Google Gemini for Q&A, quiz, summarization and learning recommendations.
Storage	No database; the supplied design states that no user data is stored.
Testing	Quiz parsing tests cover plain JSON, markdown-fenced JSON and JSON surrounded by text.
Security	Gemini API key is expected through an environment variable; the README instructs users not to commit/share .env.

Submission Note

Before submitting, ensure the actual project folder contains every file marked Yes, run the application successfully, execute the unit tests, and replace any 'Not provided' items with the real links/files if the evaluator requires them.